

CSTM: Canonical Space Transformation

Three cameras, same scene, different focal lengths

Wide (canonical)

Ultra-wide

Telephoto



$f = 14\text{mm}$ 4032×3024

$\omega = 1.71$



Enlarged $1.71 \times$

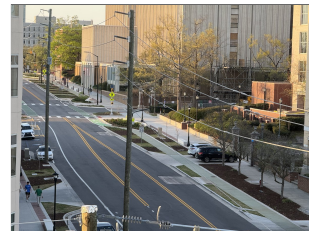


$f = f^c = 24\text{mm}$ 5712×4284

$\omega = 1.0$

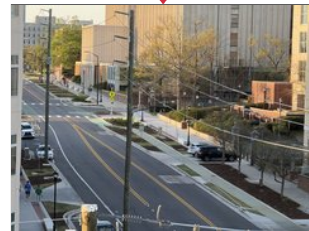


Unchanged



$f = 120\text{mm}$ 4032×3024

$\omega = 0.2$



Shrunk $0.2 \times$

CSTM-image: $I_c = \text{resize}(I, \omega)$ where $\omega = f^c/f$

All images now appear as if captured by the canonical camera ($f^c = 24\text{mm}$)

Objects at the same real-world distance occupy the same number of pixels.
The network receives consistent supervision across all cameras.